

# The Hidden Language of Cross-Species Cooperation

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In the animal kingdom, cooperation isn't limited to members of the same species. A groundbreaking review published in *Animal Behaviour* shows that communication plays a crucial role in helping different species work together. Led by Dr. Katie Dunkley at the University of Oxford, an international team examined how animals use calls, body movements, visual displays, and other signals to **synchronize** behavior and maintain **mutually beneficial** partnerships. One striking example is the greater honeyguide (Indicator indicator), which uses specialized calls to attract humans and lead them to bees' nests. The bird also responds to human calls, creating a two-way dialogue. In Mozambique, fishermen and dolphins have developed a similar partnership: dolphins signal when to cast nets, though the cues vary by region. Cleaner fish like *Labroides dimidiatus* perform distinctive dances and display bright colors to signal their role to predatory clients, allowing them to safely remove parasites. Even insects get in on the act lycaenid butterfly larvae use chemical and vibrational signals to recruit ants as bodyguards. The review emphasizes that communication helps manage risk. Signals allow animals to identify trustworthy partners and avoid exploitation. For instance, cleaner shrimp use specific movements to reassure larger fish that they won't be eaten. The researchers also explored how these **interspecies** signals evolve. Many likely began as simple cues traits that influence another animal's behavior without originally being meant as communication. Over generations, they become more specialized. Others may have been **repurposed** from behaviors used in parenting or conflict resolution. "Studying how information flows between species gives us a powerful window into how communication systems originate, change and sometimes **coevolve**," says Dunkley. The findings challenge scientists to look beyond visual signals and consider **multisensory** communication, as many species rely on chemical or vibrational cues that are easy to miss. This research not only deepens our understanding of animal behavior but also offers insights into the very roots of language.

## Word Treasure

**interspecies** -- Between different species of animals or plants.

**synchronize** -- To make actions happen at the same time or in a coordinated way.

**mutually beneficial** -- Helping both sides gain something positive from a relationship.

**coevolve** -- When two species influence each other's evolution over time because of their close interaction.

**multisensory** -- Using several different senses like sight, sound, and smell at once.

**repurposed** -- Taken from one use and adapted for a different, new purpose.

## Background you need

Interspecies cooperation, known as mutualism, is a well-documented phenomenon in biology--examples include bees pollinating flowers and oxpeckers cleaning large mammals.

The greater honeyguide (*Indicator indicator*) has been guiding human honey-hunters in sub-Saharan Africa for centuries, and indigenous communities like the Yao in Mozambique have developed specific calls to communicate with this bird.

Similarly, cooperative fishing between humans and dolphins has been observed not only in Mozambique but also in Brazil and Myanmar, suggesting that such partnerships can arise independently when both sides benefit.

Game theory, a branch of mathematics, helps scientists model how trust and signaling evolve in these partnerships.

Dr. Katie Dunkley's review, published in *Animal Behaviour*, synthesizes decades of observations to understand the hidden language of cooperation; the University of Oxford, where she works, is one of the world's oldest universities (founded in the 11th century) and a leader in biological research.

The journal *Animal Behaviour*, established in 1953, is a premier outlet for ethology, and Dunkley's work challenges traditional boundaries between human and animal communication, suggesting that the roots of language might be found in these ancient alliances.

## Break it down

**LEAD:** Broad claim that cooperation isn't limited to same species

+ STUDY: Review in *Animal Behaviour* led by Dr. Katie Dunkley examines communication across species

**EXAMPLES:** Four vivid cases of cross-species partnerships

+ Greater honeyguide: specialized calls to lead humans to bees' nests, two-way dialogue

+ Mozambique fishermen & dolphins: dolphins signal when to cast nets

+ Cleaner fish: dances and bright colors to advertise cleaning services to predators

+ Lycaenid butterfly larvae: chemical and vibrational signals to recruit ant bodyguards

**FUNCTION:** Communication helps manage risk, identify trustworthy partners

+ EXAMPLE: Cleaner shrimp use specific movements to reassure fish

**EVOLUTION:** Signals likely evolved from simple cues (unintended effects) over many generations

+ ORIGIN: Some may be repurposed from parenting or conflict behaviors

**QUOTE:** Dunkley says studying information flow reveals origins of communication

**BROADER VIEW:** Calls for multisensory research beyond visual signals

+ IMPLICATION: Insights into the roots of language

## Why it matters

Understanding how animals from different species communicate and cooperate challenges our assumptions about language and intelligence, and could inspire new approaches to collaboration in human society, from diplomacy to AI design.

## Test what you remember

**1. Which bird species guides humans to bees' nests?**

- ☐ (A) African grey parrot
- ☐ (B) Greater honeyguide
- ☐ (C) European robin
- ☐ (D) Common myna

**2. In Mozambique, what animals cooperate with fishermen to catch fish?**

- ☐ (A) Seabirds
- ☐ (B) Dolphins
- ☐ (C) Sea turtles
- ☐ (D) Octopuses

**3. What do cleaner fish like *Labroides dimidiatus* do for their clients?**

- ☐ (A) Help them hunt
- ☐ (B) Remove parasites
- ☐ (C) Guard their eggs
- ☐ (D) Lead them to food

**4. Which insect larvae recruit ants as bodyguards?**

- ☐ (A) Caterpillars of monarch butterflies
- ☐ (B) Lycaenid butterfly larvae
- ☐ (C) Ladybug larvae
- ☐ (D) Mosquito larvae

**5. According to the review, how do many interspecies signals likely begin?**

- ☐ (A) As complex songs
- ☐ (B) As random mutations
- ☐ (C) As simple cues that influence another's behavior without being intended as communication
- ☐ (D) As learned traditions passed down through generations

**6. Who led the interdisciplinary team that published this review?**

- ☐ (A) Dr. Simone Marchi
- ☐ (B) Dr. Katie Dunkley
- ☐ (C) Dr. Congyu Wu
- ☐ (D) Dr. Donald Stephens

Answer key (try the quiz first!): 1: B 2: B 3: B 4: B 5: C 6: B

## Questions to think about

- 1. Wonder of nature's teamwork:** These partnerships show that cooperation can evolve across species lines, creating remarkable relationships that benefit all participants and revealing nature's ingenuity.
- 2. Fragility of mutualisms:** Many of these cooperative systems are threatened by habitat destruction and climate change. For instance, overfishing can break the dolphin-fisherman bond, and deforestation disrupts honeyguide-human partnerships, leading to loss of cultural and ecological heritage.
- 3. Evolution of signaling:** The review highlights that communication often starts as simple, unintentional cues that gradually become specialized signals. This gradual process helps scientists understand how even rudimentary interactions can, over generations, evolve into complex languages.
- 4. Beyond visual signals:** Future studies might combine acoustic, chemical, and vibrational sensors to uncover hidden communication networks, which could lead to conservation strategies that protect not just species but their collaborative relationships.

MY THOUGHTS (at least 20 words):